

Research Timeline

Week of 3/2: Data Collection and Coding Data

Week of 3/9: Spring Break and Data Collection and Coding Data

Week 3/16: Analyzing Data and Primary Data Report (class meets online)

Research Overview

The topic of my research paper is “The Role of Machine Learning in the Data Science Community.” Through this research, I intend to examine the utilization of machine learning in my discourse community. It includes how prevalent machine learning is in this community, algorithms that are commonly used in the data science community, and the most popular topics discussed among members in this community.

The dataset that will be used in the analysis chapter is taken from the Stack Exchange forum and processed in R Studio for data cleaning and further analysis.

Research Question

The main research question of my paper is “The Role of Machine Learning in the Data Science Community.”

Text Selection

The list of questions that will be analyzed.

Top 5 by Score

Question.Id	Title	Score	View.Count	Answer.Count	Owner.Display.Name	Creation.Date	Link
1	19048 What is the difference between test set and validation...	570	675608	12	xiaohan2012	2011-11-28	https://stats.stackexchange.com/questions/19048/...
2	133656 How to understand the drawbacks of K-means	448	180054	5	KevinKim	2015-01-16	https://stats.stackexchange.com/questions/133656/...
3	99171 Why is Euclidean distance not a good metric in high d...	366	172636	8	tealeef	2014-05-18	https://stats.stackexchange.com/questions/99171/...
4	222179 How to know that your machine learning problem is h...	301	37255	3	Tim	2016-07-05	https://stats.stackexchange.com/questions/222179/...
5	7207 ROC vs precision-and-recall curves	273	139185	7	Amelio Vazquez-Reina	2011-02-14	https://stats.stackexchange.com/questions/7207/ro...

Top 5 by Date

Question.Id	Title	Score	View.Count	Answer.Count	Owner.Display.Name	Creation.Date	Link
1	136572 Use of training data that has been labeled by the AI m...	12	1617	1	raner	2025-11-03	https://datascience.stackexchange.com/questions/1/...
2	131488 In training a neural network, why don't we take the d...	12	2041	4	Leo Juhlin	2025-03-03	https://datascience.stackexchange.com/questions/1/...
3	126337 Why does data science see class imbalance as a probl...	23	1313	2	Dave	2024-01-09	https://datascience.stackexchange.com/questions/1/...
4	120764 How does an LLM "parameter" relate to a ...	18	19966	1	slim_wizard	2023-04-06	https://datascience.stackexchange.com/questions/1/...
5	590861 What is the difference between something being &qu...	33	5745	5	stochasticmrf	2022-10-02	https://stats.stackexchange.com/questions/590861/...

Comparative Analysis

Questions in the subset top 5 by score and top 5 by date are different in approach since they ask about different steps in the data science workflow (data cleaning, data wrangling, exploratory data analysis, feature engineering, predictive modeling, and data visualization).

Primary Research Methods

Which method are you using?

I plan to use the text analysis method for my research.

Why did you choose this method?

I chose this method since this method allows me to have more individuals whom I can observe and make summaries of.

How will you conduct this research?

First, I will collect the data from the Stack Exchange community forum. Then I will clean the data and explore the data through graphical summaries.

What challenges do you anticipate?

The challenges will be the time taken for me to realize all my ideas into a tangible action.

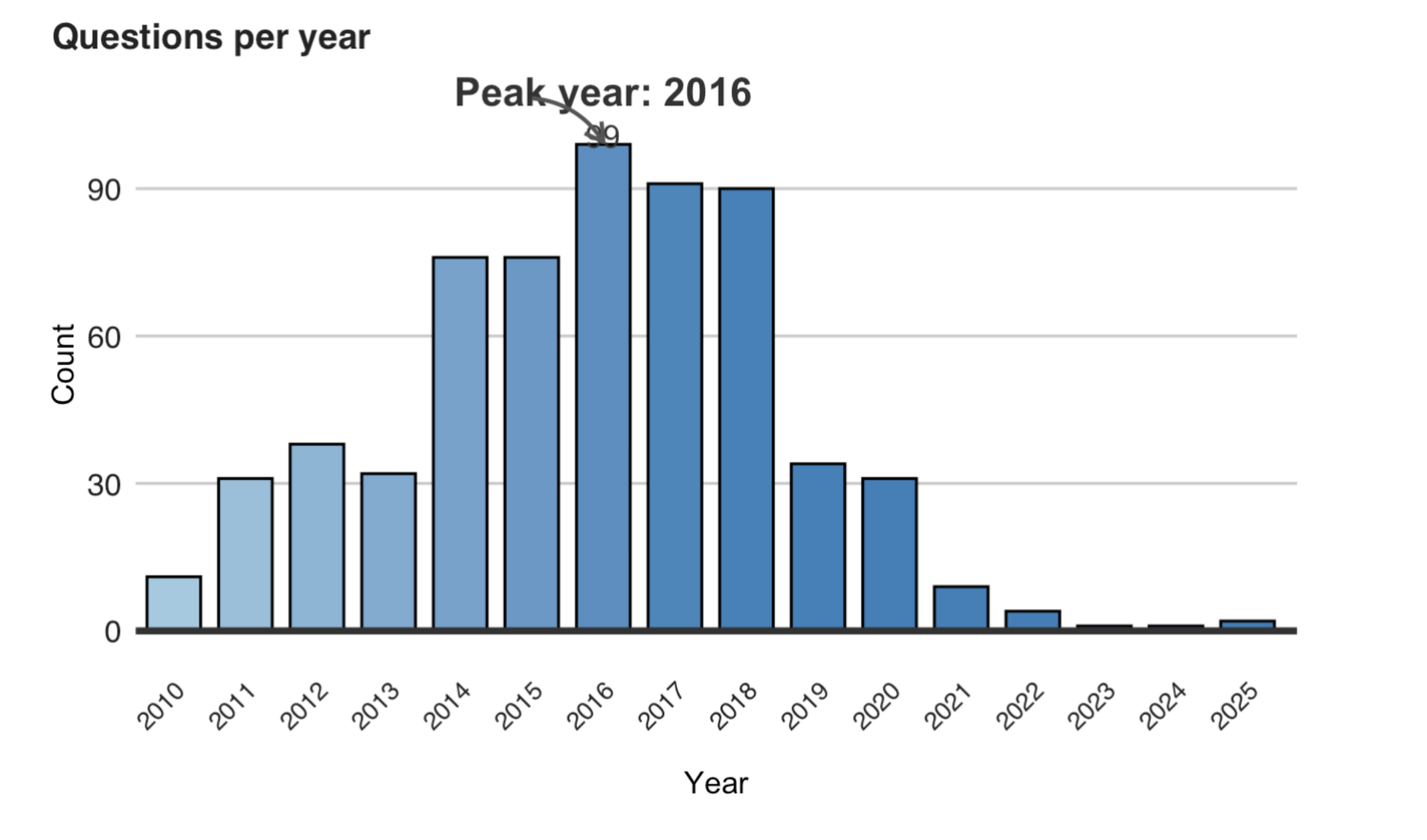
Purpose of Analysis

- Examining how machine learning concepts are discussed in a public forum.
- Assessing areas where members struggle the most.
- Identifying patterns about machine learning.

Key Features

- Model Evaluation
- Algorithm Strengths and Limitations
- High-Dimensional Data
- Statistical Terminologies
- Machine Learning Languages
- Semi-Formal

Questions per Year



Examples & Evidence

“Why is Euclidean distance not a good metric in high dimensions?”

“What is the difference between test set and validation set?”

“How to understand the drawbacks of K-means”

“ROC vs precision-and-recall curves”

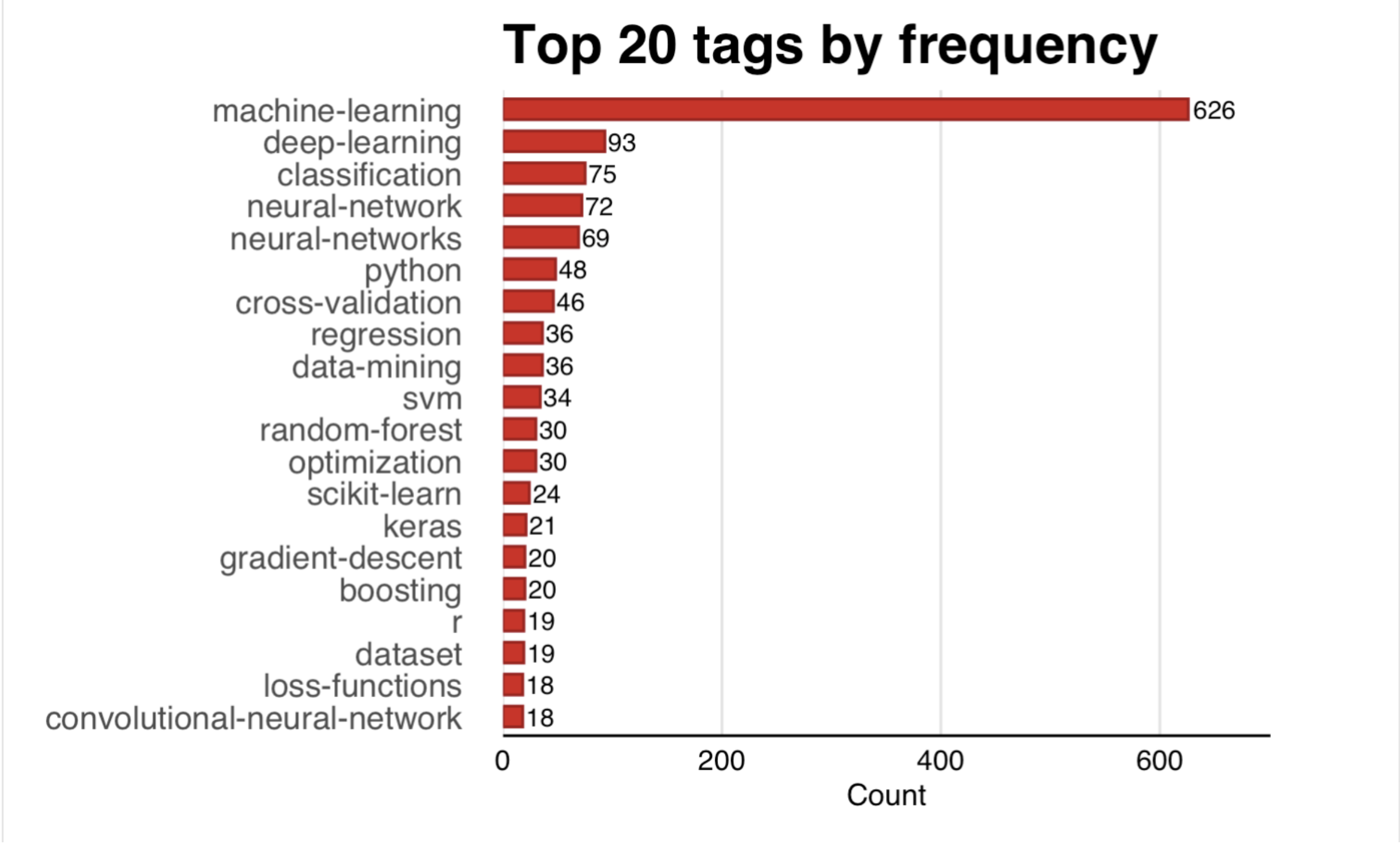
“In training a neural network, why don't we take the derivative with respect to the step size in gradient descent?”

Findings & Interpretation

- Conceptual questions were still asked
- Model Evaluation is one of the major concerns
- Interpretations matter

Machine learning is the most discussed topic among this discourse community.

Top 20 Tags



Links to Relevant Documents

<https://drive.google.com/file/d/1k6wSzzI8QDEazg8m1Dx1s5B7CyCQiwqF/view?usp=sharing>

Research Memo

Since 2016, the number of active members in this community has experienced a decrease. From 2021 to 2024 there was a sharp drop in the number of questions in the Stack Exchange forum. Additionally, Machine learning became the most widely discussed topic in the data science community.